B.M.S. COLLEGE OF ENGINEERING

AI-POWERED COLLEGE ASSISTANT FOR BMSCE

Using MERN Stack, NLP & LLM

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- MCA – 3rd Semester
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Introduction of the Project

In today’s college environment, a huge amount of data like attendance, marks, subjects, and notices is stored in different systems.
Students and teachers often struggle to find this information quickly.

Our project solves this problem by using Artificial Intelligence and a chatbot interface.
Instead of searching in portals, users can simply ask questions like:

- “What is my attendance?”
- “Show my internal marks”

The chatbot understands the question and gives a clear and simple answer.

Project Background

Colleges manage large amounts of data like attendance, marks, and notices. Students and teachers often struggle to access this information quickly using traditional portals.

Project Scope

The project provides a chatbot-based system where users can check academic data, apply for leave, and receive notices through a single intelligent platform.

Project Objectives

The main objective is to simplify academic communication by using AI to provide fast, accurate, and user-friendly access to college information.

Team Members

The project is developed by Arijeet Kumar Das and Gowtham J who designed, built, and tested the system using MERN stack, AI, and email automation.

PROBLEM STATEMENT

Current college systems:

- Are difficult to navigate
- Take time to search
- Do not support natural language
- Require manual communication for tasks like leave requests

There is a need for a smart, fast, and user-friendly system to access academic information and perform routine tasks.

PROPOSED SOLUTION

We developed an AI-Powered College Assistant that allows:

- Students to check attendance, marks, and submit leave
- Teachers to view student performance
- Admin to manage data and send notices

All through a chatbot interface.

USER ROLES

STUDENT

- Check attendance
- View marks
- Apply for leave
- View notices

TEACHER

- View student details
- Add marks and attendance
- Receive notices and leave applications

ADMIN

- Add users
- Manage subjects
- Send notices
- Receive leave applications

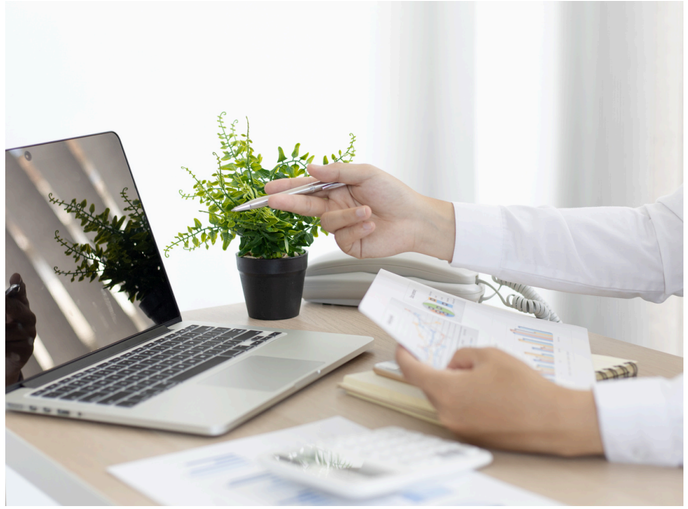
EXISTING VS PROPOSED SYSTEM

EXISTING SYSTEM

- Students must log into different portals to check marks, attendance, and notices.
- Information is scattered and takes time to find.
- Leave applications are done manually or through emails.
- No intelligent or conversational interface is available.

PROPOSED

- A single AI-powered chatbot provides all academic information.
- Students and teachers can ask questions in natural language.
- Leave applications and notices are sent automatically using email.
- The system is fast, smart, and easy to use.



SYSTEM ARCHITECTURE

- The system follows a three-tier architecture.
- The frontend provides the chatbot and user dashboards.
- The backend handles authentication, processes user requests, connects to the AI model, and manages email services.
- The database stores all academic data such as students, teachers, attendance, marks, and notices.
- All layers work together to deliver fast, secure, and accurate responses.
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DATABASE DESIGN

The system uses MongoDB with Mongoose models.
Each model represents a real college entity:

- Admin – system control
- Student – student details, marks, attendance
- Teacher – teacher information
- Subject – subject and class data
- Notice – announcements
- Chat – chatbot messages
- FAQ – common questions

This structure keeps the data organized and easy to access.

Expected Results

By using separate Mongoose models for each entity, the system becomes scalable, structured, and easy to maintain, while allowing the chatbot to return accurate academic information quickly.

SRS – Software Requirements

- Frontend: React.js
- Backend: Node.js and Express.js
- Database: MongoDB
- AI Integration: NLP and Large Language Model (Hugging Face)
- Email Service: NodeMailer
- Browser: Chrome or any modern web browser

SRS – Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 8 GB
- Storage: 20 GB free disk space
- Internet Connection
- System: Windows / Linux / Mac

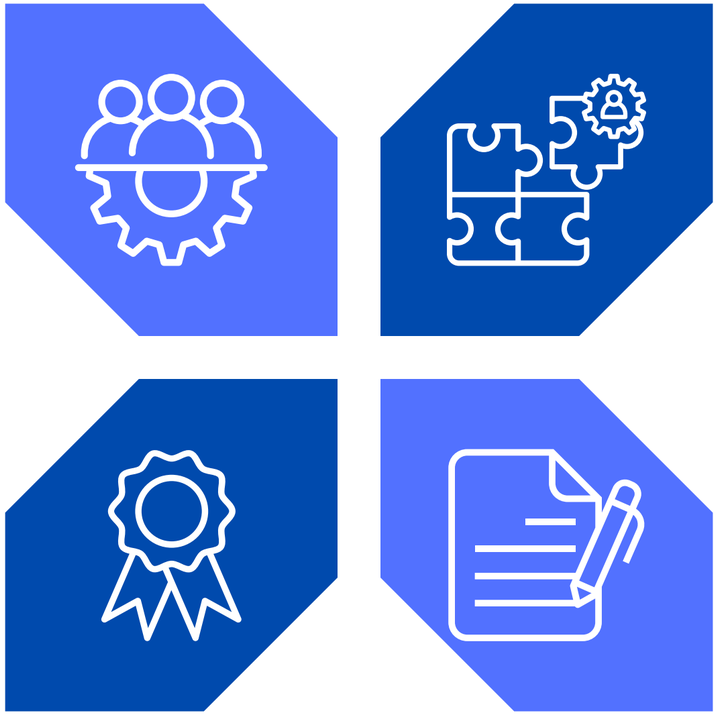
FUNCTIONAL TESTING

Functional testing checks whether the system works according to the requirements.

It verifies:

- Student login and dashboard
- Chatbot queries for attendance and marks
- Leave application submission
- Notice sending
- Role-based access (Student, Teacher, Admin)

This ensures that all features behave correctly.



NON FUNCTIONAL TESTING

Non-functional testing checks how well the system performs.

It verifies:

- System response time
- Chatbot accuracy
- Data security
- Email delivery reliability
- Stability after updates

This ensures the system is fast, secure, and reliable.

CONCLUSION

The AI-Powered College Assistant successfully provides a smart way to access academic information through a chatbot.

By combining MERN stack, AI, NLP, and email automation, the system makes communication faster, easier, and more efficient.

It improves the overall academic experience for students, teachers, and administrators.

FUTURE ENHANCEMENTS

- **Voice-Based Assistant** Students and teachers will be able to speak to the chatbot instead of typing, making the system more user- friendly.
- **Mobile Application** A mobile app can be developed so users can access the college assistant anytime from their phones.
- **Automatic Leave Approval System** The system can be enhanced to automatically approve or reject leave based on attendance and rules.
- **Advanced Analytics Dashboard** Teachers and admin can get graphs and reports of student performance, attendance trends, and usage statistics.
- **Cloud Deployment** The system can be hosted on cloud platforms to support more users and ensure high availability.

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REFERENCES

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- **React.js Documentation** https://react.dev
- **Node.js Documentation https://nodejs.org/en/docs**
- **Express.js Documentation https://expressjs.com**
- **Hugging Face (LLM API)** https://huggingface.co/docs
- **Nodemailer Documentation https://nodemailer.com**

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THANK YOU

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